

A jump-process presentation of the localized Weil form

The archimedean symbol as zero density, a broken Perron–Frobenius structure, and selection rules for a source

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Abstract

The localized Weil quadratic form on an interval I_L is usually written as an archimedean energy, a contact constant, a rank-two pole form and a sum of prime translations. We give a presentation in which the first, second and fourth of these are a single object. Adding and subtracting the value of the prime symbol at the origin puts everything but the pole term in Lévy–Khinchine form: the target is a jump Dirichlet form whose Lévy measure has the archimedean density as its continuous part and an atom of mass $2(\log p)p^{-m/2}$ at each prime period $m \log p$, together with one explicit constant. Two identities organize the account. The archimedean symbol together with the contact is exactly twice the derivative of the Riemann–Siegel theta function, so it is integration against the smooth zero-counting measure; the contact constant is not an independent object but the $-\log \pi$ inside that derivative, and the negativity of the archimedean part is confined to a band that shrinks with L and is empty above $\log 13$. Splitting the form by the sign of its symbol rather than by that constant then gives a compression presentation that moves strictly less to the subtracted side. The jump presentation carries a Perron–Frobenius structure: its resolvent is positivity preserving, as Beurling–Deny requires, and the rank-two pole form destroys that structure at every L . We explain what a cone can and cannot supply here — Krein–Rutman adds nothing to the spectral theorem for a self-adjoint form, Birkhoff bounds a spectral gap and never a spectral radius, and Collatz–Wielandt is the only cone theorem that bounds what must be bounded — and we use the part of the form that does lie in a cone to prove an elementary necessary condition: if the Weil form is positive on I_L then $\alpha_L \leq 2 \sinh(L/2) + L$, where α_L is a Perron-root deficit of an explicitly nonnegative kernel. The condition is sharp, it is bounded below by a single positive trial function, and a trial function violating it would show that the Riemann hypothesis fails; a scan of 7713 intervals finds no violation. We close with eight necessary conditions that any source for the localized form must satisfy. Nothing here establishes Weil positivity, and no statement below assumes the Riemann hypothesis.

1 Introduction

The Weil criterion makes the Riemann hypothesis the positivity of an explicit quadratic form on compactly supported test functions. Localized to an interval $I_L = (-L/2, L/2)$ it is the form

$$Q_L[f] = K[E_L f] + w_0 \|f\|^2 + P_L[f] - 2 \sum_{m \log p < L} (\log p) p^{-m/2} \operatorname{Re} \langle E_L f, U_{m \log p} E_L f \rangle, \quad (1.1)$$

an archimedean energy K , a negative contact constant w_0 , a rank-two form P_L carrying the two poles of ζ , and a sum over prime powers. The companion investigation of this program asks for a

physical system whose state norm is (1.1); the present paper asks a prior and entirely unconditional question, namely what (1.1) is.

1.1 What is proved here

The four terms of (1.1) are not four things. Section 3 shows that the first two are one: the archimedean symbol together with the contact is exactly $2\theta'(\tau)$, twice the derivative of the Riemann–Siegel theta function, so that

$$K[E_L f] + w_0 \|f\|^2 = \int_{\mathbb{R}} |\widehat{E_L f}(\tau)|^2 d\bar{N}(\tau), \quad \bar{N}(T) = \frac{\theta(T)}{\pi} + 1, \quad (1.2)$$

integration against the smooth zero-counting measure. The contact constant then requires no separate account: it is the $-\log \pi$ inside θ' . Three consequences follow at once. The archimedean part is a *signed* measure, negative exactly below $\tau_* = 6.2898\dots$, and once the prime contacts are added the negative band shrinks with L and is empty above $\log 13$; the crossing at $p = 13$ is precisely the point at which the summed prime contacts overtake $-2\theta'(0) = -w_0$; and the symbol has the graded expansion $\log(\tau/2\pi) - \frac{1}{24}\tau^{-2} - \frac{7}{960}\tau^{-4} + \dots$, with no odd orders, which is four separate conditions on a candidate rather than one.

Section 4 shows that the prime sum joins them. Adding and subtracting the value of the prime symbol at $\tau = 0$ puts everything but the pole term in Lévy–Khinchine form,

$$Q_L[f] = \mathcal{E}_L[E_L f] + \gamma_L \|f\|^2 + P_L[f], \quad (1.3)$$

where $\mathcal{E}_L$ is the jump Dirichlet form of the nonnegative Lévy measure

$$\mu_L = 2n_\gamma(r) dr + 2 \sum_{m \log p < L} (\log p) p^{-m/2} \delta_{m \log p} \quad (1.4)$$

and γ_L is an explicit constant. The archimedean density and the prime atoms are the continuous and atomic parts of one measure: a source for (1.1) is a jump process whose jumps are the primes.

Section 5 uses (1.2) to sharpen the compression presentation of the companion manuscript, splitting the form by the sign of its symbol rather than by a constant, and Section 6 takes up what (1.3) buys in the way of positivity structure. Jump forms carry Perron–Frobenius structure: their resolvents are positivity preserving. We verify that here and show that the rank-two pole form destroys it, at every L and however positive the constant is made — a third independent appearance of the pole term as the obstruction of this subject. We then say precisely what a cone can and cannot do. For a self-adjoint form Krein–Rutman adds nothing to the spectral theorem; Birkhoff’s contraction bounds a spectral gap and never a spectral radius; and Collatz–Wielandt, which does bound a spectral radius, needs a pointwise supersolution, which is exactly what the pole removes.

Section 7 applies the cone to the part of the form that is still inside one. Writing $Q_L = S_L + P_L$ with $S_L = \mathcal{E}_L + \gamma_L \|\cdot\|^2$, the form S_L is of Stieltjes type with constant diagonal, so its least eigenvalue is a Perron-root deficit $-\alpha_L$ of an explicitly nonnegative kernel. It has exactly one negative direction, and that direction is $\cosh(x/2)$ to four digits — which is the *positive* direction of the pole form. Evaluating Q_L there gives

$$Q_L \geq 0 \text{ on } I_L \implies \alpha_L \leq 2 \sinh(L/2) + L, \quad (1.5)$$

and because α_L is a Perron root, a single strictly positive trial function bounds it from below. A trial function violating (1.5) would show that the Riemann hypothesis fails. A scan of 7713 intervals finds no violation, and finds the least relative margin to be 9.5×10^{-4} .

Section 8 collects eight necessary conditions on a source, and Section 9 says what remains.

1.2 What is not claimed

Nothing here establishes Weil positivity on any interval, and no statement assumes the Riemann hypothesis. The presentations of Sections 3–5 are exact rearrangements: they do not make the problem smaller, and Section 5 says so quantitatively. The necessary condition of Section 7 is necessary and not sufficient, and Remark 7.3 gives the quantitative reason: it tests the one direction in which the difficulty is absent. Its holding over the range scanned is the expected outcome and is not evidence for the Riemann hypothesis. Every numerical statement is labelled as such, and the check programs of Appendix A state what they do and do not establish.

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2 Conventions and the target

All Hilbert inner products are conjugate-linear in the first argument. For $L > 0$ let $I_L = (-L/2, L/2)$ and let E_L denote extension by zero from I_L to $\mathbb{R}$; unless another domain is stated, $f \in C_c^\infty(I_L; \mathbb{C})$ and $F = E_L f$. The test functions carry no boundary conditions: the localization is by support alone. We use

$$\widehat{F}(\tau) = \int_{\mathbb{R}} e^{-i\tau x} F(x) dx, \quad U_d F(x) = F(x - d), \quad \|F\|^2 = \frac{1}{2\pi} \int_{\mathbb{R}} |\widehat{F}(\tau)|^2 d\tau, \quad (2.1)$$

and for $r > 0$

$$n_\gamma(r) = \frac{e^{-r/2}}{1 - e^{-2r}} = \sum_{n \geq 0} e^{-a_n r}, \quad a_n = 2n + \frac{1}{2}. \quad (2.2)$$

The archimedean energy, the contact constant and the pole form are

$$K[F] = \int_0^\infty [2\|F\|^2 - 2\operatorname{Re}\langle F, U_r F \rangle] n_\gamma(r) dr = \frac{1}{2\pi} \int_{\mathbb{R}} b(\tau^2) |\widehat{F}(\tau)|^2 d\tau, \quad (2.3)$$

$$b(\tau^2) = \sum_{n \geq 0} \frac{2}{a_n} \frac{\tau^2}{a_n^2 + \tau^2} = \operatorname{Re} \psi\left(\frac{1}{4} + i\tau/2\right) - \psi\left(\frac{1}{4}\right), \quad (2.4)$$

$$w_0 = \psi\left(\frac{1}{4}\right) - \log \pi = -\gamma_E - \frac{\pi}{2} - 3 \log 2 - \log \pi = -5.37218\dots, \quad (2.5)$$

$$P_L[f] = 2 \left| \int_{I_L} f \cosh(x/2) \right|^2 - 2 \left| \int_{I_L} f \sinh(x/2) \right|^2, \quad (2.6)$$

with $\psi = \Gamma'/\Gamma$ and γ_E Euler's constant. The target is

$$Q_L[f] = K[E_L f] + w_0 \|f\|^2 + P_L[f] - 2 \sum_{m \log p < L} (\log p) p^{-m/2} \operatorname{Re}\langle F, U_{m \log p} F \rangle, \quad (2.7)$$

the coefficient being the primitive length $\log p$ and not $m \log p$. The classical Weil criterion is that the Riemann hypothesis holds if and only if $Q_L \geq 0$ for every $L > 0$; we use it as input and do not reprove it. These are the conventions of the companion manuscript of this program [1, 2, 3], to which we refer for the program's objective and for the results quoted by number below.

Two facts from that manuscript are used repeatedly. The pole form splits by parity, $Rf(x) = f(-x)$,

$$P_L|_{\text{even}} = +2 \left| \int_{I_L} f \cosh(x/2) \right|^2, \quad P_L|_{\text{odd}} = -2 \left| \int_{I_L} f \sinh(x/2) \right|^2, \quad (2.8)$$

and the target has the spectral form

$$Q_L[f] = \sum_{\rho} \widehat{F}(\gamma_\rho) \overline{\widehat{F}(\overline{\gamma_\rho})}, \quad \gamma_\rho = -i(\rho - \tfrac{1}{2}), \quad (2.9)$$

the sum over the nontrivial zeros in symmetric order. The right side of (2.9) is very much smaller than the individual terms of (2.7); for a smooth input on I_4 the four terms are 3.40, -8.06 , 9.11 , -4.46 and their sum is 1.4×10^{-7} . This is why the assembly of Q_L from (2.7) in double precision is unusable, and why the numerical work of Appendix A either avoids the cancellation or is confined to quantities in which it does not occur.

3 The archimedean symbol is the smooth zero density

3.1 The identity

Proposition 3.1 (Density identity). *For all real τ ,*

$$b(\tau^2) + w_0 = \operatorname{Re} \psi(\tfrac{1}{4} + i\tau/2) - \log \pi = 2\theta'(\tau), \quad (3.1)$$

where $\theta(t) = \arg \Gamma(\tfrac{1}{4} + it/2) - \tfrac{t}{2} \log \pi$ is the Riemann–Siegel theta function. Consequently, with $\bar{N}(T) = \theta(T)/\pi + 1$ the smooth part of the Riemann–von Mangoldt counting function,

$$K[E_L f] + w_0 \|f\|^2 = \int_{\mathbb{R}} |\widehat{E_L f}(\tau)|^2 d\bar{N}(\tau). \quad (3.2)$$

Proof. The first equality in (3.1) is (2.4) plus (2.5). Differentiating θ gives $\theta'(t) = \tfrac{1}{2} \operatorname{Re} \psi(\tfrac{1}{4} + it/2) - \tfrac{1}{2} \log \pi$, which is the second. For (3.2), K is the Fourier multiplier with symbol $b(\tau^2)$ by (2.3) and $w_0 \|f\|^2$ is the multiplier with symbol w_0 by Plancherel, so the sum has symbol $2\theta'$; and $\bar{N}' = \theta'/\pi$, so $\tfrac{1}{2\pi} \cdot 2\theta' = \bar{N}'$. $\square$

Both sides of (3.1) are computed by unrelated routines in `check_density_symbol.py` and agree to sixteen digits.

3.2 Four consequences

The contact needs no separate account. It is a standing question of this program how a source is to produce a contact constant more negative than any of its channels. By (3.2) the constant is not a separate object: it is the $-\log \pi$ in θ' , the normalization of the zero density. A candidate whose spectral weight has the right logarithmic slope and the wrong intercept does not have a small error; it has the wrong density of states.

The archimedean form is signed, and its negative band is small. $2\theta'$ vanishes at $\tau_* = 6.28984\dots$, against $2\pi = 6.28319\dots$, and is negative below it: $\bar{N}$ decreases there. In the prime-free subtraction form of the companion manuscript the source-side symbol is

$$\sigma_L(\tau) = b(\tau^2) + w_0 + \Sigma_L = 2\theta'(\tau) + \Sigma_L, \quad \Sigma_L = \sum_{p < e^L} \frac{2 \log p}{\sqrt{p} + 1}, \quad (3.3)$$

which is negative exactly on $|\tau| < \tau_L$, with

L	0.8	1	5/4	3/2	2	5/2
τ_L	3.5504	3.5504	1.6396	1.6396	0.4745	0.2520

and no negative band at all for $L > \log 13$.

The sign change at $p = 13$ has a meaning. The constant of that presentation is $c_L = \sigma_L(0) = 2\theta'(0) + \Sigma_L$, so $c_L > 0$ is exactly $\Sigma_L > -w_0 = 5.37218\dots$: the summed prime contacts overtake the maximum negativity of the smooth zero density. That is why the crossing sits at $p = 13$ and nowhere else.

A graded condition on a candidate. Stirling for θ' gives, with every coefficient verified numerically to five digits at $\tau = 20, \dots, 10^3$,

$$b(\tau^2) + w_0 = \log \frac{|\tau|}{2\pi} + 0 \cdot |\tau|^{-1} - \frac{1}{24} |\tau|^{-2} + 0 \cdot |\tau|^{-3} - \frac{7}{960} |\tau|^{-4} + \dots \quad (3.4)$$

These are four separate falsifications rather than one: the leading logarithm with unit coefficient, which is the marginal or borderline-Gagliardo case; the intercept $-\log 2\pi$, which is the contact; the absence of the $|\tau|^{-1}$ term, which is the reflection symmetry of the form seen in the expansion; and then $-1/24$ and $-7/960$.

Remark 3.2 (Comparison with Suzuki's small-interval law). Suzuki writes the localized form as $\langle A_a v, v \rangle$ for a self-adjoint A_a on $L^2(-a, a)$ and proves, unconditionally, that its lowest eigenvalue is continuous in a and, for sufficiently small a , positive, simple and equal to $\log(1/a) + \mu_1 - \log(2\pi) + \psi(2) - 1 + O(a)$ [2]. With $a = L/2$ the leading behaviour is (3.4) evaluated at the interval's own frequency scale $\tau \sim 1/a$: the first two terms agree, the constant included. We record this because the two derivations are independent.

4 The target is a jump form

Proposition 4.1 (Jump form). *For $f \in C_c^\infty(I_L)$ and $F = E_L f$ put*

$$\mu_L = 2n_\gamma(r) dr + 2 \sum_{m \log p < L} (\log p) p^{-m/2} \delta_{m \log p}, \quad \gamma_L = w_0 - 2 \sum_{m \log p < L} (\log p) p^{-m/2}, \quad (4.1)$$

and

$$\mathcal{E}_L[F] = \int_0^\infty [\|F\|^2 - \operatorname{Re}\langle F, U_r F \rangle] d\mu_L(r). \quad (4.2)$$

Then $\mu_L \geq 0$, $\mathcal{E}_L \geq 0$, and

$$Q_L[f] = \mathcal{E}_L[E_L f] + \gamma_L \|f\|^2 + P_L[f]. \quad (4.3)$$

Proof. For the continuous part of (4.1), (4.2) is (2.3). For the atomic part, write each prime term of (2.7) as $-2c \operatorname{Re}\langle F, U_s F \rangle = -2c \|F\|^2 + 2c [\|F\|^2 - \operatorname{Re}\langle F, U_s F \rangle]$ with $c = (\log p) p^{-m/2}$ and $s = m \log p$; the first terms collect into $\gamma_L - w_0$ and the second into the atomic part of (4.2). Nonnegativity is $\|F\|^2 - \operatorname{Re}\langle F, U_r F \rangle = \frac{1}{2} \|F - U_r F\|^2$. $\square$

Equivalently, in symbols, $b(\tau^2) + w_0 - 2 \sum c \cos(\tau s) = \Phi_L(\tau) + \gamma_L$ with $\Phi_L(\tau) = \int (1 - \cos \tau r) d\mu_L(r)$ a continuous negative definite function in the sense of Schoenberg. So (4.3) is a Lévy–Khinchine representation of the whole of (2.7) except the pole term, and it is elementary; what is worth saying is what it means.

One object, not two channels. The archimedean density and the prime atoms are the continuous and atomic parts of the single nonnegative measure μ_L . A source for (2.7) is, up to the constant and the two poles, the Dirichlet form of a Lévy jump process whose jumps are the primes, with mass $2(\log p) p^{-m/2}$ at jump length $m \log p$. This matters because the companion manuscript proves that no coherent source has the archimedean energy and the summed prime references as its two channel norms: the interference bound fails at explicit test functions and by a growing factor. A presentation in which the two are one object is what that exclusion asks for, and (4.3) supplies it.

The constant. γ_L is large and negative, -6.3524 at $L = 1$ and -18.3878 at $L = 3$, with $\gamma_L = w_0 - 4e^{L/2}(1 + o(1))$ by the prime number theorem. It is a different bookkeeping from the constant c_L of (3.3), which uses the mirror contact $2dr/(1+r)$ in place of the full $2dr^n$.

A ground-state reformulation. Taking the least Rayleigh quotient of (4.3) against $\|f\|^2$ gives, exactly,

$$\lambda_{\min}(\mathcal{E}_L + P_L; \|\cdot\|^2) = -\gamma_L + \lambda_{\min}(Q_L; \|\cdot\|^2), \quad (4.4)$$

so that Weil positivity on I_L says precisely that the ground-state energy of the jump form together with the pole term is at least $2 \sum_{m \log p < L} (\log p) p^{-m/2} - w_0$. In a 48-cell basis at $L = 1$ the two sides of (4.4) are 6.354438 and 6.352442, the difference being $\lambda_{\min}(Q_L)$. This is a re-presentation and not a theorem, but it places the criterion where the spectral theory of nonlocal Dirichlet forms on an interval is the relevant apparatus.

5 Splitting by the sign of the symbol

The companion manuscript presents the target as a compression, $Q_L = K_+ - T_L$, with K_+ the archimedean energy together with the positive part of the constant c_L of (3.3) and the even half of the pole form, and T_L the negative part of that constant, the odd half of the pole form and the mirror prime references; both sides are positive term by term. That presentation moves the whole constant to whichever side its sign puts it, at every frequency. By Section 3 only the band $|\tau| < \tau_L$ has to move.

Proposition 5.1 (Symbol split). *Let $\sigma_L^\pm = \max(\pm\sigma_L, 0)$ for σ_L as in (3.3), and define on $C_c^\infty(I_L)$*

$$K_+^{\text{new}}[f] = \frac{1}{2\pi} \int_{\mathbb{R}} \sigma_L^+(\tau) |\widehat{E_L f}|^2 d\tau + 2 \left| \int_{I_L} f \cosh(x/2) \right|^2, \quad (5.1)$$

$$T_L^{\text{new}}[f] = \frac{1}{2\pi} \int_{\mathbb{R}} \sigma_L^-(\tau) |\widehat{E_L f}|^2 d\tau + 2 \left| \int_{I_L} f \sinh(x/2) \right|^2 + \sum_{p \in \mathcal{P}} \tilde{B}_{r_p, d_p}[E_L f]. \quad (5.2)$$

Then $Q_L = K_+^{\text{new}} - T_L^{\text{new}}$ identically, both sides are positive term by term, and $K_+^{\text{new}} \preceq K_+$, strictly except at $\tau = 0$.

Proof. Start from the prime-free subtraction form $Q_L = A_L - \sum_p \tilde{B}_{r_p, d_p}$ of the companion manuscript, in which A_L is the archimedean energy together with the constant c_L and the pole form. By (3.3) its first two terms are the Fourier multiplier with symbol σ_L ; split that as $\sigma_L^+ - \sigma_L^-$ and split P_L by (2.8). A multiplier with nonnegative symbol is a positive form, and each $\tilde{B}_{r_p, d_p}$ is positive. For the comparison, σ_L^- attains $(c_L)_-$ only at $\tau = 0$ because $2\theta'$ is strictly increasing on $\tau > 0$, and $K_+ - K_+^{\text{new}}$ is the multiplier with symbol $(c_L)_- - \sigma_L^- \geq 0$. $\square$

The earlier presentation is the case in which σ_L^- is replaced by the constant $(c_L)_-$ everywhere. In a 48-cell basis the least generalized eigenvalue improves as follows, both presentations assembled independently and both identities verified to machine zero.

L	4/5	1	5/4	3/2	2	5/2	3
$\min \text{spec}(Q_L; K_+)$	3.68e-4	3.68e-4	2.81e-4	2.65e-4	1.81e-4	1.89e-4	1.15e-4
$\min \text{spec}(Q_L; K_+^{\text{new}})$	1.45e-3	1.25e-3	6.84e-4	5.68e-4	2.45e-4	2.11e-4	1.15e-4
ratio	3.93	3.40	2.43	2.15	1.35	1.11	1.00

These are one-sided subspace values: above one they would certify that the domination fails, below one they certify nothing. *The gain is a constant factor and not a change of kind:* the domination remains saturated and the contraction remains critical. What the split buys is structural. The constant w_0 never appears as an independent object; for $L > \log 13$ the subtracted side is only the odd pole term and the mirror references, with no constant at all; and the band that has to move is explicit and shrinking.

6 A Perron–Frobenius structure, and what breaks it

6.1 The cone the jump form carries

Jump Dirichlet forms carry an order structure that quadratic forms in general do not. By the Beurling–Deny criteria the semigroup and resolvent of such a form are positivity preserving: if $u \geq 0$ pointwise then $(\mathcal{A} + \lambda)^{-1}u \geq 0$ for $\lambda > 0$, where $\mathcal{A}$ is the generator. In a basis of indicator cells on I_L , in which a nonnegative function is a nonnegative coefficient vector, this says that the inverse matrix has no negative entries.

That is what happens. In a 48-cell basis, the inverse of the archimedean energy plus a positive constant has *no* negative entries, at $\lambda \in \{1, 5\}$ and every L tested; and so does the inverse of the archimedean energy plus the constant c_L wherever that operator is still positive definite, which fails only at small L , where the bare archimedean form is known to be indefinite.

6.2 The pole form destroys it

Adjoining the rank-two pole form P_L removes the structure entirely.

L	4/5	1	3/2	2	3
negative fraction of $(K + 5)^{-1}$	0.000	0.000	0.000	0.000	0.000
negative fraction of $(K + 5 + P_L)^{-1}$	0.404	0.511	0.640	0.703	0.734

and the effect worsens with L . The mechanism is the Sherman–Morrison formula: adding $\sigma w w^\top$ subtracts $\sigma(A^{-1}w)(A^{-1}w)^\top / (1 + \sigma w^\top A^{-1}w)$ from the inverse, and when $A^{-1} \geq 0$ entrywise and $w \geq 0$ — as $\cosh(x/2) > 0$ is — that is a nonnegative rank-one subtraction, which drives entries negative. *Even the positive half of the pole form breaks inverse positivity.*

This is the third independent appearance of the pole term as the obstruction of this subject, after the impossibility of supplying its negative direction by an adjoined positive channel and the parity split (2.8).

6.3 What a cone can and cannot supply

It is worth being precise, because the conclusion is not that cone methods are unavailable but that three of the four standard ones cannot bound the quantity in question.

(a) The existence of an invariant cone is automatic. By Vandergraft’s theorem a matrix leaves some proper cone invariant as soon as its spectral radius is an eigenvalue, which for a self-adjoint operator is always. So the question is never whether a cone exists but whether a *usable* one, supplied by the structure, does.

(b) Krein–Rutman adds nothing to the spectral theorem here. Its content in dynamics is that a transfer operator is not self-adjoint and has no a priori spectral theory, so the existence of a leading eigenpair is a real statement. The operators of this subject are self-adjoint by construction and their leading eigenpairs exist for free.

(c) Birkhoff’s contraction bounds a gap, never a radius. A positive map taking a cone into itself with finite projective diameter Δ contracts the Hilbert projective metric with ratio $\tanh(\Delta/4)$, which controls $|\lambda_2|/|\lambda_1|$ and the convergence to the leading eigenvector. It says nothing about λ_1 , and Weil positivity is a statement about λ_1 alone.

(d) **Collatz–Wielandt does bound a radius, and needs a cone.** For a cone-positive M ,

$$\rho(M) = \inf\{\lambda : \exists u \in \text{int } C, Mu \preceq \lambda u\}, \quad (6.1)$$

so a single pointwise supersolution certifies $\rho(M) \leq 1$: it converts an operator inequality into a pointwise inequality between two explicit functions. That is what a cone is worth here, and Section 6.2 is exactly what removes it for Q_L .

The next section applies (6.1) to the largest piece of the target that is still inside a cone.

7 A necessary condition

7.1 The Stieltjes part and its Perron root

Write (4.3) as $Q_L = S_L + P_L$ with

$$S_L[f] = \mathcal{E}_L[E_L f] + \gamma_L \|f\|^2. \quad (7.1)$$

Away from the diagonal the kernel of S_L is $-\mu_L \leq 0$, and in the cell basis its diagonal is constant, so S_L is of Stieltjes type: writing $S_L = d_L I - N_L$ gives N_L symmetric, entrywise nonnegative and with zero diagonal, whence

$$\lambda_{\min}(S_L) = d_L - \rho(N_L), \quad (7.2)$$

ρ being the Perron root. Everything indefinite in the target has been moved into two explicit scalars and the rank-two P_L . By Collatz–Wielandt, for every strictly positive u ,

$$\min_i \frac{(N_L u)_i}{u_i} \leq \rho(N_L) \leq \max_i \frac{(N_L u)_i}{u_i}, \quad (7.3)$$

a two-sided bracket on $\lambda_{\min}(S_L)$ from one positive trial function, with equality at the Perron vector. At $L = 2$, where $\lambda_{\min}(S_L) = -4.3037$, the constant function gives $-4.7732 \leq \lambda_{\min}(S_L) \leq -3.8465$ from a single evaluation and no spectral computation.

7.2 One negative direction, and it is the pole's positive direction

Put $\alpha_L = -\lambda_{\min}(S_L)$. Numerically S_L has exactly one negative eigenvalue at every $L \in \{1/2, 3/4, 1, 3/2, 2, 3, 4\}$, counted by Sylvester's law; that the count is at most one is forced by positivity of Q_L , since P_L has one negative direction. Its ground state ψ has cosine $|\langle \psi, \cosh(x/2) \rangle| / \|\cosh(x/2)\|$ equal to 0.9844, 0.9918, 0.9950, 0.9979, 0.9988, 0.9997, 0.9998 at those seven values of L . By (2.8) $\cosh(x/2)$ spans the *positive* direction of P_L . So the single negative direction of S_L and the single positive direction of the pole form are the same vector to four digits. That the count is one is forced; that the directions coincide is not forced by anything proved here, and is the sharpest statement we have of what the pole term is for.

7.3 The condition

Proposition 7.1 (Necessary condition). *Let ψ be a normalized ground state of S_L , $S_L[\psi] = -\alpha_L$ with $\alpha_L > 0$. If $Q_L \geq 0$ on $C_c^\infty(I_L)$, then*

$$\alpha_L \leq 2 \left| \int_{I_L} \psi \cosh(x/2) \right|^2 \leq 2 \|\cosh(x/2)\|_{L^2(I_L)}^2 = 2 \sinh(L/2) + L. \quad (7.4)$$

Proof. By (2.8), $0 \leq Q_L[\psi] = -\alpha_L + 2|\int \psi \cosh(x/2)|^2 - 2|\int \psi \sinh(x/2)|^2$. Drop the last term and apply Cauchy–Schwarz, using $\int_{L_L} \cosh^2(x/2) dx = L/2 + \sinh(L/2)$. $\square$

L	1/2	3/4	1	3/2	2	3	4
α_L	0.9138	1.4157	1.9863	3.0994	4.3037	7.2281	11.2285
$2 \sinh(L/2) + L$	1.0052	1.5177	2.0422	3.1446	4.3504	7.2586	11.2537
relative margin	9.1e−2	6.7e−2	2.7e−2	1.4e−2	1.1e−2	4.2e−3	2.2e−3

7.4 It is a disproof test

Two facts make (7.4) operational. First, the Galerkin value of α_L in *any* finite basis is a lower bound for the true one, since a Rayleigh minimum over a subspace is at least the minimum over the space. Second, by (7.3) every strictly positive u gives

$$\alpha_L \geq \frac{1}{h} \left(\min_i \frac{(N_L u)_i}{u_i} - d_L \right), \quad (7.5)$$

requiring no optimality and no spectral computation; the constant function already gives $\alpha_L \geq 3.85$ at $L = 2$ against a threshold of 4.35. A positive u , or a finite basis, making the left side of (7.4) exceed $2 \sinh(L/2) + L$ would show that Q_L is not positive and hence that the Riemann hypothesis fails.

We have run the test. S_L is symmetric Toeplitz in the cell basis, so matrix-vector products are circulant embeddings and α_L is the deficit of the largest eigenvalue of a nonnegative operator, which is the well-conditioned end for Lanczos; a column build and eigensolve at 4000 cells takes about a fifth of a second. Convergence at $L = 1$ is 1.9873106, 1.9874609, 1.9874827, 1.9874884, 1.9874935 at $m = 250, 10^3, 4 \cdot 10^3, 1.6 \cdot 10^4, 6.4 \cdot 10^4$.

Remark 7.2 (The scan). Over 7713 values of L in $[1/2, 10]$ at 4000 cells, with every prime-power threshold sampled at $+10^{-7}$, $+10^{-3}$ and $+2 \cdot 10^{-2}$ in addition to a uniform grid, there is *no* violation of (7.4). The least absolute margin is 0.0187 at $L = 5.3396$, of relative size 9.5×10^{-4} ; the least relative margin is 3.2×10^{-4} , near $L = 10$. The absolute margin does not tend to zero: it oscillates in roughly $[0.019, 0.09]$ across the range while both sides grow like $e^{L/2}$. This is floating point with a Lanczos tolerance of 10^{-12} and a discretization error near 10^{-5} , both far below the margin, but it is not interval arithmetic and is not a certificate.

Remark 7.3 (Why it is not sufficient, quantitatively). (7.4) tests Q_L on the single direction in which S_L is negative and the pole form is positive — the easy direction, where the cancellation is rank one. On the orthogonal complement S_L is bounded below only by its second eigenvalue, while the pole form still contributes $-2|\int f \sinh(x/2)|^2$, of size up to $2 \sinh(L/2) - L$:

L	4/5	1	3/2	2	3	4
$\lambda_2(S_L)$	3.01e−2	1.22e−2	2.9e−4	2.5e−4	2.2e−4	2.4e−4
$2 \sinh(L/2) - L$	2.15e−2	4.22e−2	1.45e−1	3.50e−1	1.26	3.25
ratio	1.40	0.288	2.0e−3	7.0e−4	1.7e−4	7.4e−5

For $L \geq 3/2$ the available positivity on the complement is three to four orders of magnitude too small, so the insufficiency is quantitative rather than a missing lemma. The ratio crosses one near $L \approx 0.85$, a little above $\log 2$, which is the range in which a one-vector test is close to sufficient. Perron–Frobenius and Collatz–Wielandt are theories of the extreme eigenvalue, and nothing in them reaches λ_2 .

An exploratory computation, outside the checks, records that $\lambda_2(S_L)$ is already near 2×10^{-4} for $L \geq 3/2$, so the near-degeneracy of Q_L is largely inherited from S_L rather than produced by the pole term: the pole supplies a rank-one cancellation of one large negative eigenvalue, and does not create the criticality.

8 Selection rules for a source

A source for the localized form is a complex-linear Φ_L from $C_c^\infty(I_L)$ into a positive Hilbert space with $\|\Phi_L(f)\|^2 = Q_L[f]$. The following are necessary conditions on any such Φ_L , ordered by what they cost to apply. Conditions 1, 2 and 5 are results of the companion manuscript and are stated here for completeness; conditions 3, 4, 6, 7 and 9 are results of this paper or of the investigation accompanying it. The point of collecting them is that testing a candidate by matching the terms of (2.7) one at a time is not a weaker version of the right question: every term can be matched exactly while no source exists.

1. **Reflection.** The pairing commutes with $x \mapsto -x$: every source carries a unitary J with $J\Phi_L(f) = \Phi_L(f(-\cdot))$ and $J^2 = \text{id}$, whose eigenspaces carry the two directions of (2.8).
2. **Unboundedness.** Φ_L is unbounded relative to the input L^2 norm, because $Q_L[\chi e^{iNx}]$ grows like $\|\chi\|^2 \log N$. This excludes every source with a bounded spectral variable, and in particular the conformal Wilson pairing, whose operator has spectrum in $[-2, 2]$, so that its kernel is entire in the separation and has no atoms. The criterion is unbounded spectrum, not open versus closed geometry: the test functions carry no boundary conditions.
3. **The spectral weight is the smooth zero density**, to four orders: the leading logarithm with unit coefficient, the intercept $-\log 2\pi$, the absence of an $|\tau|^{-1}$ term, and then $-1/24$ and $-7/960$, by (3.4). A candidate with the right slope and the wrong intercept has the wrong density of states.
4. **The archimedean channel spectrum is $\{2n + \frac{1}{2}\}$ with unit residues**, exactly, by (2.2). Reading n_γ as a channel spectrum assumes an identification of the arithmetic coordinate with Euclidean time which a model must supply; where it applies, it says the transverse tower has spacing 2 and the quarter shift.
5. **Every channel decomposition passes the interference bound** $tA[f] + t^{-1}B[f] + C[f] \geq 0$ for all $t > 0$. Applied to the archimedean energy and the summed prime references this fails at the indicator of $I_{5/4}$ and by a growing factor in L . Apply it first: it is cheap and it excludes most proposals.
6. **The archimedean part is not a lifted circle weight.** A Bloch lift of a weight on S^1 has multiplier $\tau \mapsto w(e^{id\tau})$: continuous, positive, periodic and bounded, with purely atomic off-diagonal kernel. The archimedean symbol is unbounded with a smooth, nowhere vanishing off-diagonal kernel. No finite sum of such lifts, at any parameters, any number of primes and any quantization parameter, is the archimedean form.
7. **The contraction is critical, with a margin collapsing in L .** Weil positivity on I_L is a domination whose saturation $1 - \min \text{spec}(Q_L; K_+)$ is below 10^{-9} at $L = 1$ and below 10^{-24} at $L = 3$. Any hypothesis supplying a margin uniform in L — a self-map into a fixed compact subset of a cone's interior, a uniform contraction ratio, a spectral gap — proves something false.

8. **The complete identity at $L = 1$,** the only sufficient test on this list, and the expensive one.
9. **The source is a jump process with Lévy measure μ_L .** By Proposition 4.1 the entire non-pole part of the target is the Dirichlet form of (4.1): continuous part $2n_\gamma$, and an atom of mass $2(\log p)p^{-m/2}$ at each $m \log p$. This subsumes conditions 3 and 4 and replaces them by one, and it removes the appearance of two mechanisms: the archimedean and prime halves are the continuous and atomic parts of one measure.

Conditions considered and discarded are worth naming too. A ghost or indefinite sector for the pole term is not required and cannot exist in a source compatible in L , which is unitarily the evaluation map on the zeros in a positive ℓ^2 ; what survives of that discussion is condition 1. The prime-free archimedean inequality is implied by $Q_L \geq 0$ and so cannot fail unless the Riemann hypothesis does, and discriminates nothing.

9 What remains

The gap of the jump form. The second eigenvalue of S_L sits near 2×10^{-4} and is roughly constant over $3/2 \leq L \leq 4$. That is a spectral gap statement about an explicit jump Dirichlet form on an interval, it is where the criticality of the target originates, and it is well posed. An explanation of it would explain the collapse of the margin mechanically rather than measuring it. The spectral theory of nonlocal Dirichlet forms is a developed subject and has not been applied here.

The coincidence of Section 7.2. The ground state of an operator assembled from the archimedean density and the prime atoms is $\cosh(x/2)$ to four digits, and $\cosh(x/2)$ is the pole direction. Nothing proved here forces that. A proof would make the rank-one cancellation of the pole term structural rather than numerical, and would very likely sharpen Proposition 7.1.

The sharp form of the necessary condition. P_L has rank two, so the exact criterion, rather than the relaxation (7.4), is a condition on the two scalars $\langle \cosh(x/2), S_L^{-1} \cosh(x/2) \rangle$ and $\langle \sinh(x/2), S_L^{-1} \sinh(x/2) \rangle$. It is exact and small, but S_L^{-1} is dominated by the near-null direction of Remark 7.3, so the two scalars are delicate in the same way the original problem is; this relocates the difficulty rather than reducing it.

A cone adapted to the pole directions. Section 6 closes the pointwise cone and gives the reason. It does not close the possibility of a cone adapted to the two pole directions, nor the complex-cone theory appropriate to indefinite forms. The prize is named precisely: a Collatz–Wielandt pointwise supersolution.

Scope. None of the above is proposed as a route to the Riemann hypothesis. The results of this paper are unconditional structure and one necessary condition; what they are for is to constrain and to exclude candidate sources cheaply, which is the use the companion investigation makes of them.

Preparation

This working manuscript was produced with a large language model, Anthropic’s Claude (model `claude-opus-5`), working directly in the investigation repository: the identities and proofs of Sections 3–7, the check programs of Appendix A, and the drafting throughout. The research notes accompanying the investigation name the model used for each, and record the exploratory

computations that are deliberately outside the registered checks. Every statement here is classified in the text as a written proof, a one-sided numerical value, or a labelled floating-point computation.

A Checks and reproduction

Four programs accompany this paper. They use Python’s standard library only, print a JSON record to standard output, and do not overwrite the preserved records in `numerics/records/`. Counts refer to finite test cases and not to independent theorems.

- `check_density_symbol.py` (77 cases). The series and digamma forms of b with an explicit tail bound; the contact constant against its closed form; (3.1) against a central difference of $\arg \Gamma$, the two sides computed by unrelated routines; the four coefficients of (3.4); the smooth count against three published zero ordinates; and the sign structure of σ_L , including τ_* , the prime ladder and the crossing at $p = 13$, and the band τ_L .
- `check_symbol_split.py` (32,295 cases). The closed-form Toeplitz archimedean energy against direct quadrature of (2.3), agreeing to 10^{-14} ; both compression identities as finite matrix identities at seven values of L ; Cholesky certificates that the subtracted form and the negative-part multiplier are positive and that the new source side is smaller; and the one-sided saturation quotients of Section 5.
- `check_levy_dirichlet.py` (11,555 cases). (4.3) as a matrix identity at five values of L ; the jump structure of $\mathcal{E}_L$; Beurling–Deny cone positivity of the resolvent; its destruction by the pole term; and the ground-state identity (4.4).
- `check_collatz_wielandt.py` (70 cases). That S_L is a Stieltjes operator with constant diagonal and N_L entrywise nonnegative; (7.2); that S_L has exactly one negative direction at seven values of L ; the overlap of its ground state with $\cosh(x/2)$; two-sided brackets (7.3) from three trial functions; and (7.4) with its margins.

Run from the investigation directory:

```
python3 numerics/check_density_symbol.py
python3 numerics/check_symbol_split.py
python3 numerics/check_levy_dirichlet.py
python3 numerics/check_collatz_wielandt.py
python3 validation/drafts.py check --replay
```

Three computations reported in the text are *not* among these programs and are recorded in the investigation’s notes: the smooth-basis saturation values quoted in selection rule 7, which use arbitrary-precision arithmetic and a truncated sum over zero ordinates; the scan of Remark 7.2, which uses a Toeplitz eigensolver; and the values of $\lambda_2(S_L)$ in Remark 7.3. Each is labelled where it appears.

The saturation quotients are **one sided**: they are Rayleigh quotients in a step or polynomial subspace, so a value above one would certify that a domination fails, while a value below one certifies nothing. The Galerkin values of α_L are the opposite way round and this is what makes Section 7.4 a test: a minimum over a subspace is at least the minimum over the space, so every computed α_L is a lower bound for the true one.

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